DS614 Big Data Engineering (Winter ’2026)

Instructor: Ankush Chander, 3209, Faculty Block-3
Credit Structure (L-T-P-Cr): 3-0-2-4
Course Placement: M.Sc Data Science, Semester-III

Course content:

This course aims to equip students with a deep understanding of the architectural and theoretical foundations of big data systems, moving from storage internals (LSM-Trees, B-Trees) to modern Data Lakehouse architectures. It establishes a rigorous framework for distributed computing, examining replication, partitioning, and consistency models (CAP, PACELC) required for fault tolerance. Students will develop engineering proficiency in designing orchestrated batch and streaming pipelines while mastering probabilistic algorithms (HyperLogLog, Count-Min Sketch) for large-scale approximation. Finally, the course emphasizes reliability engineering, inculcating best practices for data quality, schema governance, and observability in production environments.

Course uses lab and project work to ensure the balance between theory and practice, ideas and their application.

Suggested Textbook/references:

1. **Designing Data Intensive Applications** by Martin Kleppmann
2. **Fundamentals of Data Engineering** by Joe Reis and Matt Housley

Evaluation plan:

1. 2 examinations(mid sem and end sem): 40%
2. Lab submissions and Project work: 60%

Course Outcomes:
At the end of the course, students will be able to:

- **Analyze** complex distributed storage requirements to select optimal data architectures (Lakehouse vs. Warehouse) based on trade-offs between consistency, availability, and latency.
- **Design** scalable data ingestion and orchestration systems that integrate batch and streaming data (Kafka/Airflow) to meet specified throughput and latency constraints.
- **Apply** mathematical principles of distributed computing and probabilistic algorithms (HyperLogLog, Count-Min Sketch) to solve large-scale cardinality and frequency estimation problems.
- **Evaluate** the impact of partitioning, replication, and concurrency control strategies on system fault tolerance and linearizability in distributed environments.
- **Develop** automated data governance and reliability mechanisms (schema contracts, observability) that address professional responsibilities regarding data quality and integrity.

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X						X			

Tentative Lecture Distribution

Sl. No.	Description		No. of Lectures
1	**Introduction & Lifecycle**		**2**
	1.1	Data Engineering vs Data Science vs Software Eng.	1
	1.2	The DE Lifecycle: Generation, Storage, Ingestion, Transformation, Serving	1
2	**Data storage**		**8**
	2.1	Data Structures of the databases- Hash Indexes, LSM trees, B-trees	3
	2.2	Architectures: Data Warehouse vs Data Lake vs Data Lakehouse	2
	2.3	Table Formats: Bringing ACID to Data Lakes (Iceberg/Delta concepts)	1
	2.4	Column-oriented Storage & Compression techniques (Parquet/ORC)	2
3	**Ingestion & Orchestration**		**7**
	3.1	Patterns: ETL vs ELT, Change Data Capture (CDC)	2
	3.2	Stream Ingestion & Message Queues (Kafka theory: Topics, Partitions, Offsets)	2
	3.3	Orchestration: Directed Acyclic Graphs (DAGs), Idempotency, and Scheduling	2
	3.4	Encoding formats: Language-Specific Formats, JSON, XML, Binary variants, Protocol buffers etc	1
4	**Data Transformation**		**6**
	4.1	Data Modeling: Dimensional Modeling (Star/Snowflake) vs NoSQL patterns	2
	4.2	Distributed Compute Models: MapReduce Paradigm vs DAG execution (Spark)	2
	4.3	Streaming: Windowing, Watermarking, and State Management	2
5	**Reliability and Quality**		**3**
	5.1	Data Quality: Testing data, Circuit breakers, Data lineage	1
	5.2	Observability: Monitoring pipelines, SLA/SLOs	1
	5.3	Serving Layers: REST APIs vs High-throughput exports	1
6	**Distributed Systems Theory**		**8**

Sl. No.	Description		No. of Lectures
	6.1	Replication: Single-leader, Multi-leader, Quorums	2
	6.2	Partitioning: Sharding strategies, Rebalancing, Consistent Hashing	2
	6.3	Transactions - ACID, isolation levels, serializability	2
	6.4	Fault-tolerant distributed systems	2
7	**Big data algorithms**		**7**
	7.1	Approximate Counting and Morris' Algorithm	2
	7.2	Distinct Elements Problem - Count-Min Sketch	2
	7.3	Large scale Linear Algebra: Dimensionality reduction (SVD/PCA via Power Iteration)	3
Total Lectures			**41**